

# Mapping the Narrow Corridor with Large Language Models

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## Abstract

Acemoglu and Robinson’s *The Narrow Corridor* traces national histories as paths through a two-dimensional (state power, society power) space but plots none: quantifying the two axes at each moment is the labor-intensive, judgment-laden task that has kept the framework qualitative. We ask whether large language models (LLMs) can operationalize it. We introduce a reproducible, provider-agnostic pipeline that scores a country period by period using chain-of-thought (events first, then score) and in-context anchoring on prior periods, against an explicit rubric with schema-validated output. As a proof of concept we produce a six-country trajectory atlas (Iran, France, the United Kingdom, the United States, China, Chile) and compare four models. We assess the scores for consistency with an expert index (V-Dem) and for inter-model agreement — reading agreement as concurrent consistency rather than accuracy, since such indices likely appear in the models’ training data — and discuss conceptually how the biases of an LLM annotator and a human expert differ. Code, prompts, runs, and an interactive gallery of the animated trajectories are released.

## 1 Introduction

In *The Narrow Corridor* [2], Acemoglu and Robinson propose that liberty is neither the natural product of a strong state nor of a strong society, but of a contested balance between the two. They picture a country’s history as a path through a plane whose horizontal axis is the power of civil society and whose vertical axis is the power of the state; a “narrow corridor” running between the axes marks the region where the two powers grow together and liberty is sustained. The book is rich in narrative but contains *no plotted trajectories*. The obstacle is not conceptual but operational: turning centuries of a nation’s history into a sequence of (society, state) coordinates requires an enormous amount of expert judgment, and any such coding is vulnerable to the biases, selective readings, and coarse simplifications that histo-

rians rightly distrust.

This is a measurement problem, and measurement problems in the social sciences increasingly admit machine assistance. Large language models have been shown to annotate and classify political text at or above the level of crowd workers [11], and a growing literature examines their use as instruments in computational social science [19]. We ask a narrower, concrete question: *can an LLM place a country in the Narrow Corridor space, period by period, in a way that is reproducible, checkable against an established expert index, and useful to a researcher?*

We take an applied, digital-humanities stance. Our primary artifact is a **six-country trajectory atlas** — Iran, France, the United Kingdom, the United States, China, and Chile — rendered as time-shaded paths through the corridor, together with the cross-country patterns those paths reveal. The method and its validation are the backing that makes the atlas trustworthy rather than decorative. Concretely, we contribute:

1. **A scoring method.** A period-by-period pipeline combining an anchored 0–10 rubric, chain-of-thought elicitation of events and trends [18], and in-context anchoring on prior periods [5, 8].
2. **A provider-agnostic, open implementation.** Any model reachable through a unified interface can be swapped by changing one string; we release code, prompts, caches, and every run.
3. **A trajectory atlas and cross-country reading** of six countries.
4. **An interactive, publicly hosted web gallery.** A static site (GitHub Pages) presenting every trajectory as a time-shaded plot with *click-to-play animations* that reveal the path period by period and annotate the historical event driving each move; readers explore and compare models directly in the browser, turning a static figure into a navigable atlas.
5. **An assessment of the scores** for consistency with an expert index (V-Dem) and for inter-model

reliability, with explicit treatment of the training-data-overlap threat that makes such agreement concurrent consistency rather than proof of accuracy.

6. **A conceptual discussion of measurement bias:** what an LLM annotator plausibly gets right and wrong relative to a human researcher or an expert panel doing the identical task (we run no human study here; that comparison is left to future work).

## 2 Related Work

**The Narrow Corridor and measuring state/society power.** The (state, society) framing extends the institutional program of Acemoglu and Robinson [1, 2]. Quantifying the two axes connects to decades of effort to measure regime characteristics and state capacity: the Varieties of Democracy (V-Dem) project aggregates expert codings into latent indices including a core civil-society index and measures of state administrative capacity [6, 16]; Polity5 codes regime authority on an autocracy–democracy scale [14]; and Freedom House rates political rights and civil liberties [10]. These projects are themselves large expert coding operations, which makes them both our natural validation targets (Section 5.3) and our natural comparison point when we ask what an LLM changes about the coding process (Section 6).

**Text as data and LLMs as social-science instruments.** The use of automated methods to extract measures from text is well established [12]. Recent work reports that instruction-tuned LLMs match or exceed crowd annotators on political text-labeling tasks [11] and surveys their broader promise and pitfalls for computational social science [19]. A related strand uses LLMs to *simulate* human populations [3] and debates whether they can stand in for human participants at all [7]. Our task differs from labeling or simulation: we ask the model to produce a continuous, temporally coherent *measurement* of a theoretical construct, then check that measurement against independent data.

**Prompting.** Our pipeline relies on two now-standard techniques. Chain-of-thought prompting [18] improves reasoning by eliciting intermediate steps; we use it to force the model to enumerate historical events and trends *before* scoring. In-context learning [5, 8] lets the model condition on examples provided at inference time; we feed all prior periods’ scores forward so the trajectory is temporally coherent rather than a sequence of independent guesses. This anchoring is a deliberate design

choice with a cost — an early mis-score can propagate — which we return to in Section 7.

**Bias and reliability.** LLMs inherit biases from training data [4] and exhibit measurable political leanings: they do not reflect a representative cross-section of opinion [17], can carry the political slant of their pretraining corpora into downstream behavior [9], and have been probed directly for partisan bias [15]. These findings motivate both our reliability analysis — borrowing the logic of inter-coder reliability from content analysis [13] — and our discussion of how LLM bias compares to the well-known biases of human expert coders.

## 3 Method

**The space.** We place a country at each time period  $t$  at a point  $(\text{soc}_t, \text{sta}_t) \in [0, 10]^2$ , where  $\text{soc}_t$  is the power of civil society and  $\text{sta}_t$  is the power of the state. A period is a fixed span of years (e.g., a decade). We read a country as *in the corridor* in period  $t$  when its state and society power are close (near the diagonal) and *above/below* it when one clearly dominates. The dashed lines in our plots mark this band *illustratively*, near the diagonal, rather than as a fitted threshold, and are drawn at constant width — a deliberate simplification of the book’s *widening* corridor; positions are therefore read qualitatively, not from a numeric cutoff.

**Anchored rubric.** A free-floating request for “two numbers” is not a measurement. Every scoring prompt embeds a fixed 0–10 rubric that defines each band of state power (from 0–2, collapsed/nonexistent, to 9–10, overwhelming penetration of society) and society power (from 0–2, atomized, to 9–10, society pervasively checks the state), plus explicit neutrality guidance: judge each period on its own terms, weigh evidence for and against a shift, avoid presentism, and treat the two axes as independent. The rubric is what makes scores comparable across periods, countries, and models.

**Chain-of-thought: events then scores.** For each period the model is first asked to list the major events and trends affecting state or society power, separating slow-moving trends from discrete events and noting the direction of each effect. This narrative is then inserted into a second prompt that requests the numeric score, so the number is conditioned on explicit reasoning rather than produced in one leap [18].

**In-context learning: temporal anchoring.** Starting from the earliest period (scored on absolute terms), we accumulate every prior period’s (soc, sta) pair and feed the running trajectory into each subsequent scoring prompt [5]. The model reports both the change

from the previous period and the resulting absolute values, which discourages the sequence from drifting and keeps a quiet period near zero change.

**Structured, validated output.** Each scoring call returns a JSON object — the two power values, their per-period changes, a short key event, and a one-sentence justification — validated against a fixed schema. A malformed response triggers one reminder retry and then a numeric fallback extractor before an error is raised.

**Provider-agnostic access and caching.** All model calls go through a single unified interface, so a model is selected by one identifier string and providers can be swapped without code changes. Because the pipeline makes one or two calls per period and is therefore expensive, every response is cached on disk keyed by (model, prompt); re-runs and visualization iteration are free, and the released cache makes our exact runs reproducible.

**Visualization.** A run renders as (i) a static plot of society vs. state power with the path time-shaded from early (dark) to late (bright) and a year colorbar, the largest moves annotated with their year and key event; and (ii) an animation that reveals the path period by period, drawing an arrow for the move each event caused and captioning it. Figure 1 is built from the static renderings; the animations are best viewed in the interactive gallery at [https://esonghori.github.io/visualize\\_narrow\\_corridor\\_with\\_ai/](https://esonghori.github.io/visualize_narrow_corridor_with_ai/).

## 4 Experimental Setup

**Countries.** We study six countries chosen to span regions of the corridor and of the world: **Iran (Persia)**, **France**, the **United Kingdom**, the **United States**, **China**, and **Chile**. Each is scored from a country-appropriate start year through 2020 (a common cutoff for consistency across cases) in decade (10-year) periods, giving 14–24 periods per country (Table 1 lists windows and counts). Six countries is a proof of concept, not a basis for cross-national generalization.

**Models.** We compare four models from four families (Google, Anthropic, OpenAI, Alibaba): three closed-weight models (gemini-3.5-flash, claude-opus-4-8, gpt-4o) and one open-weight model (qwen-2.5-72b-instruct). All are driven through the same interface with identical prompts, isolating model effects from prompt effects.

**Consistency check against an expert index.** We compare the LLM scores to V-Dem [6], aligning each period’s midpoint year to the V-Dem year. We map LLM *society power* to the V-Dem core civil-society index (v2xcs\_ccsi) and LLM *state power* to a state-authority measure (V-Dem state authority over territory, v2terr) — a narrow, territorial-control proxy for the rubric’s broader penetration-of-society notion of state power. The mapping is fixed before inspecting correlations, and alternative capacity indicators are easy to swap in the released code. We report rank (Spearman) correlations per country on both *levels* and *first differences* (period-to-period change): levels are dominated by a shared secular trend and inflate correlation, so first differences are the more honest agreement signal. Crucially, V-Dem is itself almost certainly in the models’ training data, so we read this as *concurrent consistency with expert consensus*, not as external proof of accuracy (Section 7). Polity5 and Freedom House are natural additional anchors but are out of scope here.

## 5 Results

### 5.1 Trajectory atlas

Figure 1 shows the six trajectories under claude-opus-4-8, chosen as a single reference model for legibility; the per-model panels for all four models are in the online gallery, and their disagreement is quantified in Section 5.4. **[One paragraph reading the atlas: which countries sit inside vs. above vs. below the corridor, and the shape of the paths (e.g., loops indicating oscillation, long diagonal runs indicating co-growth).]**

### 5.2 Cross-country patterns

**[Report patterns visible across the atlas: e.g., whether corridor-resident countries (candidate: UK, France late) show tighter co-growth than outside-corridor countries (candidate: China high-state/low-society; pre-modern cases low-state), and how sharp transitions (revolutions, coups) appear as large single-period jumps.]**

### 5.3 Consistency with V-Dem

Table 1 reports correlations between the LLM scores and V-Dem over overlapping years, on both levels and first differences. **[Summarize: society vs. v2xcs\_ccsi and state vs. v2terr; contrast the (high) level correlations against the (lower, more informative) first-difference correlations; note where consistency holds vs. breaks down.]** As noted above, this measures agreement with expert con-



Figure 1: Trajectory atlas: society power (x) vs. state power (y) for six countries under `claude-opus-4-8`. Paths are time-shaded (dark = early, bright = late); dashed lines mark the corridor illustratively. Animated versions are available in the online gallery. [\[Regenerate with paper/experiments/run\\_experiments.py.\]](#)

Table 1: Consistency with V-Dem (`claude-opus-4-8`): Spearman  $\rho$  at period-midpoint years, reported as level/ $\Delta$  (levels vs. first differences). Society vs. `v2xcs_ccsi`; state vs. `v2terr`. First differences are the more honest signal.

| Country        | Window    | $N$ | Society $\rho$ | State $\rho$ |
|----------------|-----------|-----|----------------|--------------|
| Iran (Persia)  | 1880–2020 | [–] | [–] / [–]      | [–] / [–]    |
| France         | 1789–2020 | [–] | [–] / [–]      | [–] / [–]    |
| United Kingdom | 1789–2020 | [–] | [–] / [–]      | [–] / [–]    |
| United States  | 1789–2020 | [–] | [–] / [–]      | [–] / [–]    |
| China          | 1880–2020 | [–] | [–] / [–]      | [–] / [–]    |
| Chile          | 1818–2020 | [–] | [–] / [–]      | [–] / [–]    |

sensus the models were likely trained on, not independent accuracy.

#### 5.4 Model comparison and reliability

Treating each model as a “rater” in the sense of inter-coder reliability [13], Table 2 reports agreement between models on the same country. We report Krippendorff’s  $\alpha$  (interval, treating the 0–10 scores as interval-scaled and the four models as raters) on the period-to-period *changes* rather than on levels (for the same reason first differences are used in Section 5.3): agreement on changes tests whether models see the same *moves*, not merely the same secular trend. **[Summarize: which axis is more reliable, which model pairs agree most, and whether the open-weight model diverges from the closed frontier models.]**

Table 2: Inter-model reliability: Krippendorff’s  $\alpha$  (interval) on period-to-period *changes* across models, per country. Higher = models see the same moves.

| Country        | #models | Society $\alpha$ | State $\alpha$ |
|----------------|---------|------------------|----------------|
| Iran (Persia)  | 0       | [–]              | [–]            |
| France         | 0       | [–]              | [–]            |
| United Kingdom | 0       | [–]              | [–]            |
| United States  | 0       | [–]              | [–]            |
| China          | 0       | [–]              | [–]            |
| Chile          | 0       | [–]              | [–]            |
| <b>Overall</b> | –       | [–]              | [–]            |

#### 5.5 Qualitative case studies

**[Two or three short reads checking that trajectories match documented history: Iran around 1979 (sharp state consolidation / society suppression after the revolution); Chile 1973 and 1990 (coup and redemocratization); and a third (e.g., UK gradual co-growth, or China post-1949). Note where the model matches consensus and where it is anachronistic or flattens contested episodes.]**

## 6 Discussion: LLM vs. Human Researcher or Expert Panel

Both an LLM and a human expert (or panel) performing this coding are biased; the useful question is *how the bias profiles differ*, and which task properties favor which

coder.

**What a human panel brings.** Domain expertise, sensitivity to contested historiography, the ability to flag “this case is genuinely disputed,” and accountable, attributable judgment. Established indices such as V-Dem manage individual coder bias with multiple experts per case and a latent measurement model that estimates and corrects for coder disagreement [16]; content analysis formalizes this as inter-coder reliability [13]. The costs are equally real: expert panels are slow, expensive, hard to scale to arbitrary country-periods, and carry their own systematic biases — national, ideological, disciplinary — that are difficult to audit because they live in individual judgment.

**What an LLM changes.** An LLM annotator is fast, cheap at scale, and *transparent in its instructions*: the rubric and prompts are explicit artifacts that can be inspected, criticized, versioned, and re-run identically. Its bias, however, is *opaque in origin*: it reflects the distribution of its training data [4, 9, 17], which over-represents English-language, recent, and Western sources, and it tends toward received consensus, likely flattening contested or under-documented episodes rather than flagging them. It has no accountable author, and its scores can shift with model version or provider (Section 5.4).

**Reliability as the bridge.** The inter-coder logic that human panels use for humans applies to models: agreement across independent models is a measurable proxy for reliability. Where independent models agree on the moves, a score is more likely to reflect a shared signal than one model’s idiosyncrasy; where they disagree, the disagreement usefully localizes the contested cases a human panel would also flag. Reliability is necessary but not sufficient for validity, though — models trained on overlapping corpora can agree on a shared bias — which is why we treat both inter-model agreement and V-Dem consistency as evidence about *reliability and consensus*, not accuracy.

**When to use which.** We argue the LLM pipeline is best positioned as a *scalable first pass and a reproducible baseline* — generating candidate trajectories over many countries cheaply, surfacing where models disagree, and freeing expert attention for the contested cases — rather than as a replacement for expert coding where accountability and contested-case judgment are paramount.

## 7 Limitations

Several limitations bound what this proof of concept establishes. *Circularity of the consistency check.* V-

Dem and similar indices are almost certainly in the models’ training data, so agreement with them cannot distinguish measuring the construct from recalling memorized values; we therefore frame the check as concurrent consistency, not accuracy, and disentangling the two (e.g., counterfactual-history probes, or ablating the events step) is future work. *Construct compression.* The two axes reduce vast, contested constructs to single numbers, and our single state-capacity indicator only partially aligns with the theory’s notion of state power. *Coverage and language bias.* Training corpora over-represent English-language, recent, and Western sources — weakest for exactly three of our cases (Iran, China, Chile), and we prompt in English throughout. *Path dependence.* In-context anchoring can propagate an early mis-score through the whole trajectory; we do not ablate it here. *Single-model atlas.* Figure 1 uses one model for legibility; readers should treat it as one reading, with cross-model spread in Section 5.4. *Scale and statistics.* Six countries and short per-country series preclude cross-national generalization or strong significance claims. We deliberately keep the study small; the natural next steps — a human expert-coder baseline, prompt/period-granularity ablations, a non-English prompting probe, more indices, and an ensemble atlas — are left to future work.

## 8 Conclusion

We presented, as a proof of concept, a method and a provider-agnostic tool that operationalize the Narrow Corridor’s (state, society) space into reproducible trajectories, together with a protocol for checking their consistency with expert consensus and their reliability across models, and released it as a one-command exercise. The resulting six-country atlas is a first quantified rendering of a framework that has been purely qualitative, and the accompanying consistency check and bias discussion frame honestly what such LLM-generated measurements are — and are not — good for.

## Reproducibility

All code, prompts, the on-disk response cache, and every run (JSON, static plots, animations) are released at [https://github.com/esonghori/visualize\\_narrow\\_corridor\\_with\\_ai](https://github.com/esonghori/visualize_narrow_corridor_with_ai). The exact experiments in this paper are reproduced by `paper/experiments/run_experiments.py` and `analysis.py`; see `paper/README.md`. The interactive gallery of animated trajectories is hosted at [https://esonghori.github.io/visualize\\_narrow\\_corridor\\_with\\_ai/](https://esonghori.github.io/visualize_narrow_corridor_with_ai/).



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